Memorize or Average?

An Empirical Study of How Symmetric Label Noise
Degrades Seven Standard Classifiers

Independent Computational Study

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Abstract

Real-world training labels are rarely perfect, yet introductory machine-learning practice usually assumes they are. We measure how seven widely used classifiers (logistic regression, an RBF support vector machine, 1-nearest-neighbour, 15-nearest-neighbour, a decision tree, a random forest, and Gaussian naive Bayes) degrade when a fraction $\eta \in \{0, 0.1, 0.2, 0.3, 0.4\}$ of training labels is randomly corrupted, evaluating on clean test data over 20 random splits of three benchmark datasets. The classifiers fall into two clear groups. *Memorizers* (1-NN and an unpruned decision tree), whose prediction for a test point depends on a single training label, lose accuracy almost linearly with η; a one-line probabilistic model predicts their noisy accuracy from their clean accuracy with a mean absolute error of 1.3 percentage points. *Averagers* (large-k nearest neighbours, random forests, the SVM) remain close to their clean accuracy until the true class stops being a clear majority among nearby noisy labels, which explains why the binary task is the hardest at $\eta = 0.4$ and the ten-class task the easiest. Two further findings are practical: the optimal regularization strength for logistic regression shifts by one to two orders of magnitude under heavy noise, and Gaussian naive Bayes, which is highly robust on two datasets, collapses on the third because of near-zero variance estimates, a failure that a larger variance-smoothing term repairs.

Keywords: label noise; robustness; classification; k-nearest neighbours; random forests; naive Bayes; regularization

1 Introduction

Supervised learning assumes that training labels are correct. In practice they often are not: labels come from crowd workers, from automated heuristics, or from experts who disagree with one another. Northcutt et al. (2021) estimated that even widely used benchmark test sets contain a non-trivial share of label errors. A practitioner therefore needs to know not only which classifier is most accurate on clean data, but which one degrades most gracefully when the labels are imperfect.

Label noise has a long history in learning theory, going back to the random classification noise model of Angluin and Laird (1988), and it has been surveyed extensively (Frénay and Verleysen, 2014). The topic gained new attention when Zhang et al. (2017) showed that modern deep networks can perfectly fit training data with completely random labels, i.e., they can *memorize* noise. Much of the recent literature focuses on deep learning and on specialized remedies such as robust loss functions (Ghosh et al., 2017), loss correction (Natarajan et al., 2013), or filtering suspected mislabelled examples (Brodley and Friedl, 1999).

The goal of this paper is more modest and, we believe, useful for practitioners and students: to measure carefully how the *standard* classifiers found in every machine-learning library behave under the simplest noise model, and to explain the differences with simple arguments that can be checked against data. Our contributions are:

1. A controlled comparison of seven classifiers on three datasets at five noise levels, with 20 repetitions per setting.
2. A closed-form "memorization" model that predicts the noisy accuracy of 1-NN and decision trees from their clean accuracy alone, and a "plurality margin" argument that explains the behaviour of averaging methods.
3. Two ablations showing how the choice of a single hyperparameter (k in k-NN, C in logistic regression) controls robustness.
4. A diagnosis and fix for a failure mode of Gaussian naive Bayes under label noise.

2 Background and Noise Model

Symmetric label noise. Let $y \in \{1, \dots, K\}$ be the true label. Under *symmetric* (uniform) noise with rate η, a label is replaced with probability η by one of the other $K-1$ classes chosen uniformly at random. In the taxonomy of Frénay and Verleysen (2014), this is noise that is completely at random: the corruption does not depend on the features or on the true class. It is the most commonly studied benchmark setting, though real noise is often class-dependent or instance-dependent.

The plurality margin. Under symmetric noise, among the training examples that truly belong to class c, a fraction $1-\eta$ keep label c and each other class receives a fraction $\eta/(K-1)$. The advantage of the correct label over any single wrong label is therefore

$$m(\eta, K) = (1-\eta) - \frac{\eta}{K-1}. \tag{1}$$

As long as $m > 0$, the true class remains the most frequent label in any region dominated by that class, so a method that averages many labels can still recover it. The margin depends strongly on K: at $\eta = 0.4$ it equals 0.20 for $K = 2$, 0.40 for $K = 3$, and 0.56 for $K = 10$. Binary problems are thus the most fragile under symmetric noise, and become impossible at $\eta = 0.5$.

A memorization model. Consider instead a classifier whose prediction for a test point is simply the label of one training example, for instance its nearest neighbour or the single training point in its leaf. Suppose that on clean labels the classifier is correct with probability a_0, and that the noise on the retrieved training label is independent of whether that clean prediction would have been correct. If the clean prediction is right, it stays right with probability $1-\eta$. If it is wrong, the noise turns it into the correct class with probability $\eta/(K-1)$. Hence

$$a(\eta) \approx a_0(1-\eta) + (1-a_0)\,\frac{\eta}{K-1}. \tag{2}$$

For a perfect memorizer, accuracy falls linearly in η with slope close to $-a_0$. Averaging methods should do considerably better, as long as the margin in Eq. (1) is comfortably positive.

3 Experimental Setup

Datasets. We use three standard benchmark datasets distributed with scikit-learn (Pedregosa et al., 2011), originally from the UCI repository (Dua and Graff, 2019). They span binary, three-class, and ten-class problems (Table 1).

Table 1: Datasets used in the study.

Dataset	Samples	Features	Classes K
Breast Cancer (Wisconsin Diagnostic)	569	30	2
Wine	178	13	3
Digits (8×8 handwritten)	1,797	64	10

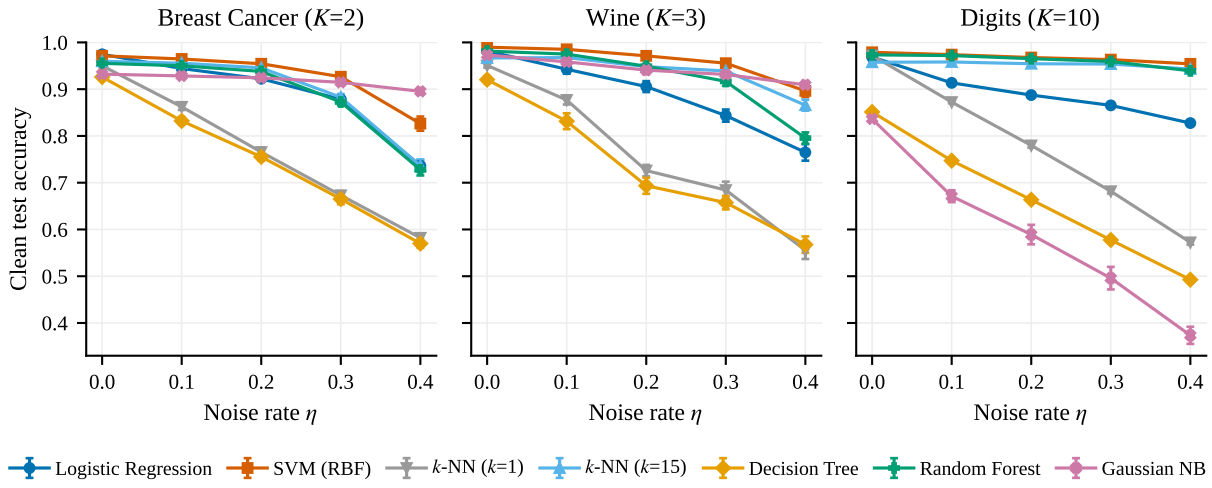


Figure 1: Clean test accuracy versus training-label noise rate η. Points are means over 20 splits; error bars show ± 1 standard error.

Classifiers. We use seven scikit-learn classifiers with library defaults unless stated: logistic regression ($C = 1$); an SVM with RBF kernel ($C = 1$, $\gamma =$ "scale"); k-nearest neighbours with $k = 1$ (Cover and Hart, 1967) and $k = 15$; an unpruned decision tree; a random forest with 200 trees (Breiman, 2001); and Gaussian naive Bayes. Logistic regression, the SVM, and k-NN are preceded by feature standardization fitted on the training set.

Protocol. For each dataset we draw 20 stratified 70/30 train–test splits. For each split and each $\eta \in \{0, 0.1, 0.2, 0.3, 0.4\}$, we corrupt exactly $\mathrm{round}(\eta\, n_{\mathrm{train}})$ training labels chosen at random, assigning each a uniformly chosen incorrect class. The *test labels are never corrupted*, so we measure how well each method recovers the true concept from noisy supervision. We report mean test accuracy over the 20 splits with its standard error. Because the training sets of different splits overlap, these standard errors somewhat understate the true uncertainty (Nadeau and Bengio, 2003); we therefore focus on differences that are large relative to them.

4 Results

4.1 Main comparison

Figure 1 shows clean test accuracy as a function of the noise rate, and Table 2 summarizes accuracy at $\eta = 0$ and $\eta = 0.4$.

Without noise, the classifiers are tightly bunched: on every dataset except Digits, all seven reach between 92% and 99% accuracy. Noise separates them sharply. Three patterns stand out.

Table 2: Test accuracy without noise and at $\eta = 0.4$ (standard error in parentheses). "Retained" is the ratio of accuracy at $\eta = 0.4$ to accuracy at $\eta = 0$, averaged over the three datasets.

	Breast Cancer		Wine		Digits		
Classifier	$\eta = 0$	$\eta = 0.4$	$\eta = 0$	$\eta = 0.4$	$\eta = 0$	$\eta = 0.4$	Retained
Logistic Regression	0.975	0.736 (0.012)	0.981	0.765 (0.018)	0.969	0.828 (0.004)	0.80
SVM (RBF)	0.971	0.826 (0.015)	0.990	0.896 (0.013)	0.979	0.954 (0.002)	0.91
k-NN (k=1)	0.950	0.582 (0.008)	0.951	0.556 (0.019)	0.972	0.572 (0.005)	0.60
k-NN (k=15)	0.960	0.736 (0.013)	0.967	0.866 (0.012)	0.958	0.943 (0.003)	0.88
Decision Tree	0.926	0.570 (0.009)	0.920	0.568 (0.017)	0.851	0.493 (0.006)	0.60
Random Forest	0.955	0.727 (0.012)	0.981	0.795 (0.012)	0.973	0.939 (0.002)	0.85
Gaussian NB	0.932	0.895 (0.007)	0.973	0.909 (0.008)	0.836	0.374 (0.018)	0.78

First, 1-NN and the decision tree lose accuracy almost linearly and retain only about 60% of their clean accuracy at $\eta = 0.4$ on every dataset. On Breast Cancer, 1-NN falls from 0.950 to 0.582.

Second, the SVM, 15-NN, and the random forest are remarkably robust at low and moderate noise. On Digits they lose less than four percentage points even at $\eta = 0.4$; the SVM keeps 0.954. On the binary Breast Cancer task, however, even these methods drop to between 0.73 and 0.83 at $\eta = 0.4$.

Third, Gaussian naive Bayes behaves inconsistently. On Breast Cancer and Wine it is the *most* robust method at $\eta = 0.4$ (0.895 and 0.909), yet on Digits it is the least robust, falling to 0.374. We return to this in Section 4.5.

Logistic regression sits between the two groups. It degrades steadily and substantially more than the SVM, retaining 80% of its accuracy on average. Section 4.4 shows that most of this gap is due to its default regularization strength rather than an intrinsic weakness.

4.2 Memorizers follow the memorization model

Figure 2(a) compares the observed accuracy of 1-NN and the decision tree with the prediction of Eq. (2), computed from each method's own clean accuracy a_0 on the same dataset. Across all 24 combinations (2 classifiers $\times$ 3 datasets $\times$ 4 noise levels), the mean absolute error is 0.013. It is 0.011 for 1-NN and 0.014 for the decision tree, and the largest single error is 0.051 (decision tree, Wine, $\eta = 0.2$).

The close agreement indicates that these two classifiers do essentially nothing to resist label noise: each corrupted training label is reproduced in the region of input space it controls. The small systematic tendency of the observed values to fall slightly *below* the prediction (mean signed error -0.010) is plausibly explained by the fact that a corrupted label can also affect neighbouring regions. For the tree, a mislabelled point creates extra splits that alter predictions near it; for 1-NN, the Voronoi cell of a mislabelled point can absorb test points that would otherwise have been classified correctly.

4.3 Averagers follow the plurality margin

For averaging methods, Eq. (1) predicts that robustness should depend on the number of classes. This is what we observe. At $\eta = 0.4$, 15-NN retains 77% of its clean accuracy on the binary task ($m = 0.20$), 90% on Wine ($m = 0.40$), and 98% on Digits ($m = 0.56$). The random forest and the SVM show the same ordering. Figure 2(b) pools all datasets and noise levels: every averaging method retains at least 93% of its clean accuracy whenever $m \geq 0.55$, but retention falls to between 76% and 85% at $m = 0.2$ (the binary task at $\eta = 0.4$). Points from different datasets with similar margins have broadly similar retention, which suggests that the margin, rather than the dataset itself, is the main driver. The ten-class problem, which one might

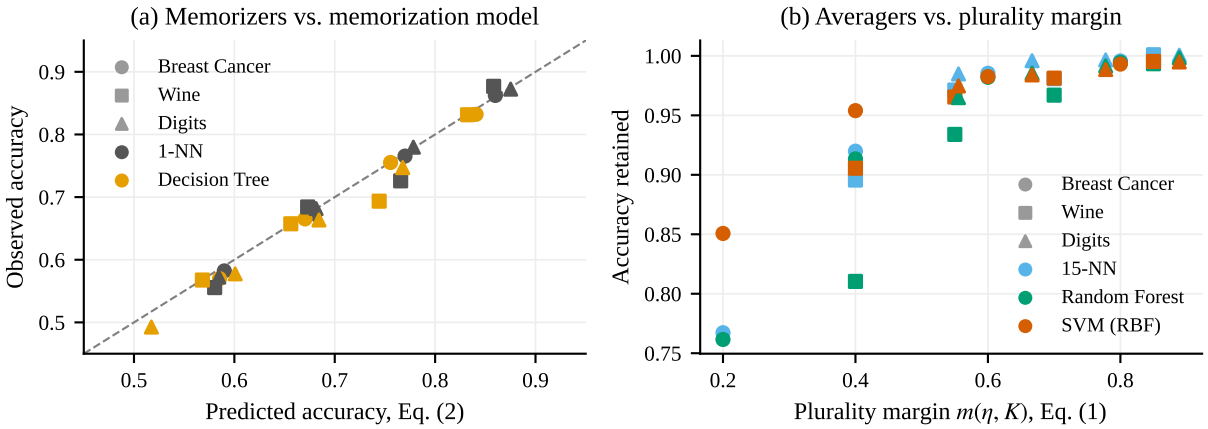


Figure 2: Two simple models explain most of the degradation. (a) Observed accuracy of 1-NN and the decision tree at $\eta \in \{0.1, 0.2, 0.3, 0.4\}$ versus the prediction of the memorization model, Eq. (2); the dashed line is $y = x$. (b) Fraction of clean accuracy retained by the three averaging methods versus the plurality margin, Eq. (1), pooling all datasets and noise levels.

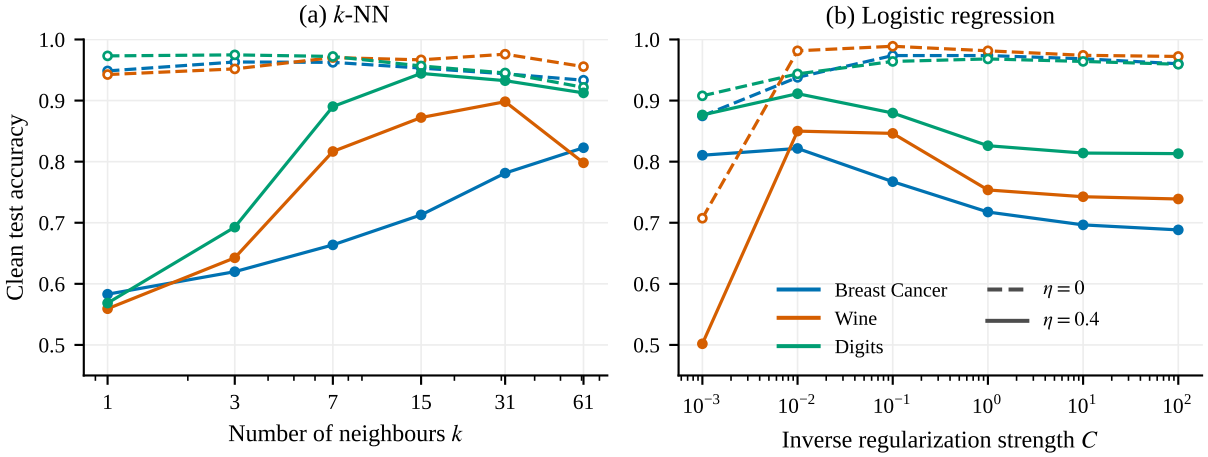


Figure 3: Effect of a single hyperparameter on robustness. Dashed lines with open markers: clean labels; solid lines: $\eta = 0.4$. (a) k-NN with varying k. (b) Logistic regression with varying C (smaller C means stronger ℓ_2 regularization).

expect to be the most difficult because it has the most classes, is the most forgiving under symmetric noise, because wrong labels are spread thinly over nine alternatives and never form a competing majority.

4.4 Ablations: one hyperparameter controls robustness

To isolate the effect of averaging, we vary k for k-NN and the inverse regularization strength C for logistic regression (10 splits each, $\eta \in \{0, 0.4\}$). Figure 3 shows the results.

Number of neighbours. On clean data, accuracy is nearly flat in k (for example, 0.943 to 0.976 on Wine across all k). At $\eta = 0.4$, accuracy rises steeply with k: on Digits from 0.569 at $k = 1$ to 0.944 at $k = 15$, and

on Breast Cancer from 0.583 at $k = 1$ to 0.823 at $k = 61$. On Breast Cancer and Digits, the best k under noise (61 and 15) is far larger than the best k on clean data (3–7 and 3); on Wine the two coincide at $k = 31$. This is the averaging mechanism in its purest form: a larger neighbourhood contains more labels, and by the law of large numbers the plurality vote converges to the true class whenever $m > 0$. Very large k eventually hurts (Wine, $k = 61$, when the neighbourhood covers about half the training set), as bias from neighbouring classes takes over.

Regularization. For logistic regression, the best C on clean data is 0.1 to 1, but under $\eta = 0.4$ it is 0.01 on all three datasets. Choosing $C = 0.01$ instead of the default $C = 1$ raises accuracy at $\eta = 0.4$ from 0.718 to 0.822 on Breast Cancer, from 0.754 to 0.850 on Wine, and from 0.826 to 0.911 on Digits. Strong regularization limits how far the decision boundary can bend to accommodate mislabelled points, which plays a role similar to averaging. The convex logistic loss is known to be sensitive to random label noise in general (Long and Servedio, 2010; Ghosh et al., 2017), but these results show that, in the low-dimensional regime studied here, careful tuning of C recovers much of the lost accuracy. A practical implication is that hyperparameters tuned on clean validation data may be poorly suited to noisy training data.

4.5 Why Gaussian naive Bayes collapses on Digits

Gaussian naive Bayes estimates a mean and variance for every feature in every class. Many Digits features are border pixels that are almost always zero, so their within-class variances can be extremely small; scikit-learn's default smoothing adds only 10^{-9} times the largest feature variance. Because the Gaussian log-likelihood contains the term $-\frac{1}{2}\sum_j \log \sigma_{cj}^2$, a class that happens to retain many near-zero variances receives a very large bonus for any test point that is zero on those pixels, which is most of them.

Label noise changes which classes keep these degenerate variances: when a corrupted example with a non-zero value in such a pixel is added to a class, that class's variance for the pixel jumps. The result is erratic. In a representative split at $\eta = 0.4$, 478 of the 540 test predictions (89%) went to just three classes (0, 5, and 7), which were precisely the three classes with the smallest summed log-variance. Two checks support this explanation. First, a nearest-centroid classifier, which uses the same noisy class means but no variances, keeps 0.887 accuracy at $\eta = 0.4$. Second, increasing the variance-smoothing parameter largely repairs the problem (Table 3). With smoothing 10^{-1}, accuracy at $\eta = 0.4$ rises from 0.369 to 0.877, and clean accuracy also improves, from 0.835 to 0.919.

Table 3: Gaussian naive Bayes on Digits with different variance-smoothing values (mean accuracy over 10 splits).

Variance smoothing	$\eta = 0$	$\eta = 0.1$	$\eta = 0.2$	$\eta = 0.3$	$\eta = 0.4$
10^{-9} (default)	0.835	0.666	0.637	0.512	0.369
10^{-3}	0.902	0.893	0.869	0.844	0.801
10^{-2}	0.915	0.906	0.886	0.870	0.858
10^{-1}	0.919	0.913	0.899	0.892	0.877

On Breast Cancer and Wine, whose features are continuous measurements without degenerate variances, naive Bayes behaves like an averager: its class means and variances are pooled over many examples, and symmetric noise mainly shrinks the class means toward each other without reversing their order.

5 Discussion

A simple taxonomy. Our results suggest a useful mental model for practitioners. A classifier whose prediction depends on *one* training label inherits the noise rate almost one-for-one. A classifier that pools *many* labels is protected by the plurality margin, and its robustness is controlled by how much pooling it does, through k, regularization strength, ensemble averaging, or smoothing. The random forest is an instructive case: each of its trees is a memorizer, yet averaging 200 bootstrap-trained trees makes the ensemble behave like an averager.

Practical recommendations. If labels may be noisy: (i) avoid 1-NN and single unpruned trees; (ii) increase the amount of averaging relative to what clean-data tuning suggests (larger k, smaller C); (iii) be most cautious with binary problems, where the plurality margin is smallest; and (iv) check variance-based models such as Gaussian naive Bayes for near-degenerate features before relying on them.

Limitations. We studied only symmetric noise. Real-world noise is often class-dependent (for example, confusing visually similar digits) or instance-dependent (ambiguous examples are mislabelled more often), and such noise can be considerably more harmful because it can create a consistent wrong majority (Frénay and Verleysen, 2014). Our datasets are small and tabular, so results may not transfer to high-dimensional data or deep networks, which have different memorization dynamics (Zhang et al., 2017). Finally, we did not tune hyperparameters separately for each noise level in the main comparison; the ablations show that doing so would narrow the gaps between some methods.

6 Conclusion

Under symmetric label noise, the main factor that determines how a standard classifier degrades is whether it memorizes or averages. Memorizers lose accuracy in proportion to the noise rate, in close agreement with a one-line model. Averagers remain robust until the true class loses its majority among nearby labels, which makes binary problems the most fragile. Robustness can often be bought cheaply by increasing the amount of averaging or regularization, but defaults tuned for clean data can be badly misplaced when labels are noisy.

Reproducibility and disclosure. All experiments used scikit-learn 1.9 with fixed random seeds for data splits and noise injection; the full study runs in a few minutes on a laptop-class CPU. The experiment code and the text of this report were prepared with the assistance of an AI system (Claude, Anthropic); all reported numbers come directly from the executed code.

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